// Day 14 of 30

VISUAL EXPLAINER

# What Happens When You Call an API

→ Swipe through all 6 slides

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# Step 1: DNS Resolution

Browser finds the server

```
You type: api.example.com/users/1

Browser asks DNS:
  'Where is api.example.com?'

DNS responds:
  '203.0.113.42'

Now browser knows the IP address.
```

💡 DNS = phonebook of the internet

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# Step 2: TCP + TLS Handshake

Establishing a secure connection

```
TCP 3-way handshake:
  Client → SYN → Server
  Server → SYN-ACK → Client
  Client → ACK → Server

TLS (HTTPS):
  → Exchange certificates
  → Agree on encryption
  → Secure channel ready
```

💡 This takes ~50-100ms. Happens once per connection.

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## Step 3: HTTP Request Sent

Your request hits the wire

```
GET /users/1 HTTP/1.1
Host: api.example.com
Authorization: Bearer eyJhbG ...
Content-Type: application/json
Accept: application/json
```

💡 Headers carry auth, format, and metadata

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# Step 4: Load Balancer → Server

Traffic routing

```
Request hits Load Balancer
  → Checks which server is healthy
  → Routes to Server 2 (least busy)

Server 2 receives request
  → Nginx/Gateway passes to Spring Boot
  → DispatcherServlet takes over
```

💡 Load balancer = traffic cop for your servers

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## Step 5: Spring Boot Processing

Controller → Service → Repository

```
DispatcherServlet
  → Filter Chain (auth, CORS)
  → @GetMapping("/users/{id}")
  → UserController.getUser(1)
  → UserService.findById(1)
  → UserRepository.findById(1)
  → Hibernate → SQL → MySQL
  → Result → DTO → JSON
```

💡 This is where YOUR code lives

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## Step 6: Response Back

Server → Client in milliseconds

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "id": 1,
  "name": "Shubham",
  "role": "Backend Engineer"
}

Total time: ~120ms
  DNS: 20ms, TCP: 30ms
  Server: 50ms, Transfer: 20ms
```

💡 Understanding this = understanding backend engineering

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```
done = true
```

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